The Institutional Stance

Julian Jara-Ettinger, Yale University, USA,

julian.jara.ettinger@yale.edu, https://compdevlab.yale.edu/

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Yarrow Dunham, Yale University, USA,

yarrow.dunham@yale.edu, https://www.socialcogdev.com/

Abstract. Human success in constructing and navigating a complex social world is typically associated with the ability to interpret other people's behavior in terms of mental states—a *mentalistic stance*. We argue that humans are endowed with a second equally powerful intuitive theory: an *institutional stance* that interprets social interactions in terms of structured institutional representations. In contrast to the mentalistic stance, which helps us predict and understand unconstrained behavior via unobserved mental states, the institutional stance shapes and regulates behavior via normative expectations embedded in roles (i.e., institutions). Precursors of the institutional stance may also help explain non-human social behavior, supporting social interaction between conspecifics that lack complex models of each others' mental states. At the same time, the institutional stance is unique in humans in its generative capacity, supporting rapid construction, tracking, and inference of institutional structures that shape our social world and enable the possibility of large scale social coordination and new forms of institutional life. Uniquely-human social cognition is best understood as an interplay between mentalistic mechanisms for predicting others' behavior embedded in normative institutional structures that help render behavior predictable.

Keywords*:* Social cognition, Coordination, Institutional Stance, Theory of Mind, roles, scripts

1. Introduction

Humans are unparalleled in the ubiquity and complexity of their social lives. From passing interactions to life-long relationships, we not only effortlessly navigate the social world, but also shape it by building social relations, communities, and societies. How do humans reach levels of social complexity that no other species has achieved? Theories of human social intelligence have been dominated by the idea that human social interactions are supported by a capacity to *mentalize*: We can represent other people's minds and attribute mental states like preferences, knowledge, and desires. This allows us to explain and predict behavior, determine how to interact with others, and even plan for how to change minds. The complexity of this *mentalistic stance* is distinctively human (Martin & Santos, 2016): it emerges early in infancy (Sodian, 2011; Onishi & Baillargeon, 2005); is supported by specialized neural circuitry (Saxe & Kanwisher, 2003; Saxe, 2006); and permeates social thought, from language understanding to moral reasoning (Jara-Ettinger & Rubio-Fernandez, 2021; Sperber & Wilson, 1986; Young et al., 2007).

Here we argue that such a view is an incomplete characterization of human social thought. We propose that human social intelligence is best understood as the co-evolution of two systems of social understanding, the *mentalistic stance* just discussed, and an *institutional stance*. This institutional stance is on equal footing with the mentalistic stance in terms of representational complexity, evolutionary history, and everyday use, but it differs in that its core representational substrate does not invoke representations of mental states. Instead, the institutional stance grounds social interaction in representations of social structures that specify how agents interact with one another as a function of relative roles with rule-like normative entailments.

Consider two simple examples that illustrate the institutional stance at work. Imagine that you are at an office building when you see Kate step out of a glass office, wave at Eric who is sitting at a desk outside, and rattle off "Can you feed my parking meter, pick up lunch for me, print the documents I just emailed you, and push my next appointment by 30 minutes?" which Eric jots down and rushes out to complete. In principle, we could interpret this event through a mentalistic stance: Kate *wants* lunch delivered, which she can eat while reviewing some documents, and she *believes* that she can fulfill her desires by uttering them to Eric. Conversely, Eric has a recursive *desire* to fulfill Kate's desires, and *believes* that he can do so by completing the tasks that Kate described. Suppose now that, over the weekend, Kate goes to the movies and spots Eric sitting a few rows behind her, apparently on a date. Kate waves him over, asks him to get her some popcorn and a fountain drink, and turns back to continue watching the movie. Under the mentalistic analysis we just applied, this interaction should be wholly unsurprising. It provides nothing more than confirmatory evidence of the mental states we already attributed to Kate. And we should further expect Eric to excuse himself and happily go fulfill Kate's desires. Why then does Kate's behavior strike us as surprising and improper, and why do we expect Eric to be similarly taken aback? We propose that this surprise emerged because we originally interpreted Eric and Kate's interaction not through a mentalistic stance but through an institutional stance, where each person had different social affordances based on the roles they occupied at that moment (e.g., Eric is Kate's assistant)—roles they do not presently occupy and that therefore no longer constrain or predict their behavior in the ways they did before.

Consider a second example. Imagine that Laurie lands in a new city and goes to a rental car agency where Paul greets her and helps her fill out the required paperwork. As Laurie is filling out the forms, she notices that the final fee is higher than she expected and starts arguing with Paul. After a minute, Paul briefly excuses himself, and a few minutes later, Alex shows up and picks up the discussion with Laurie right where Paul left off. Under the mentalistic stance, the transfer of Paul's beliefs, desires, and intentions to Alex is strikingly odd, and we should reasonably expect Laurie to be surprised about the seamless swap of communicative partners. But the uninterrupted flow of the interaction is unsurprising under an institutional stance, where we recognize that a new person occupying the same role will take on the goals and behaviors attributed to that role (i.e., placating a customer). These inferences uniquely follow from an institutional stance. To see why, consider how different our expectations would be in a situation where the institutional stance is *not* warranted. For example, imagine Laurie's surprise if, during a personal argument with a close friend, the friend excused themselves only to have a casual acquaintance step in and pick up the argument where it had been left off. In this case a seemingly similar transfer of goals would be surprising, if not off-putting.

These examples illustrate how role-based representations allow us to understand and predict behavior, and they point to commonplace social situations where mentalistic representations alone struggle to provide accurate explanations. While it may be possible to provide a mentalistic interpretation of some of these cases, doing so requires assuming increasingly complex, auxiliary, context-sensitive, and intuitively implausible mental-state attributions. By contrast, we propose that these same social interactions are seamlessly explained by appealing to an *institutional stance*, an intuitive theory for analyzing behavior in terms of underlying role-based social structures.

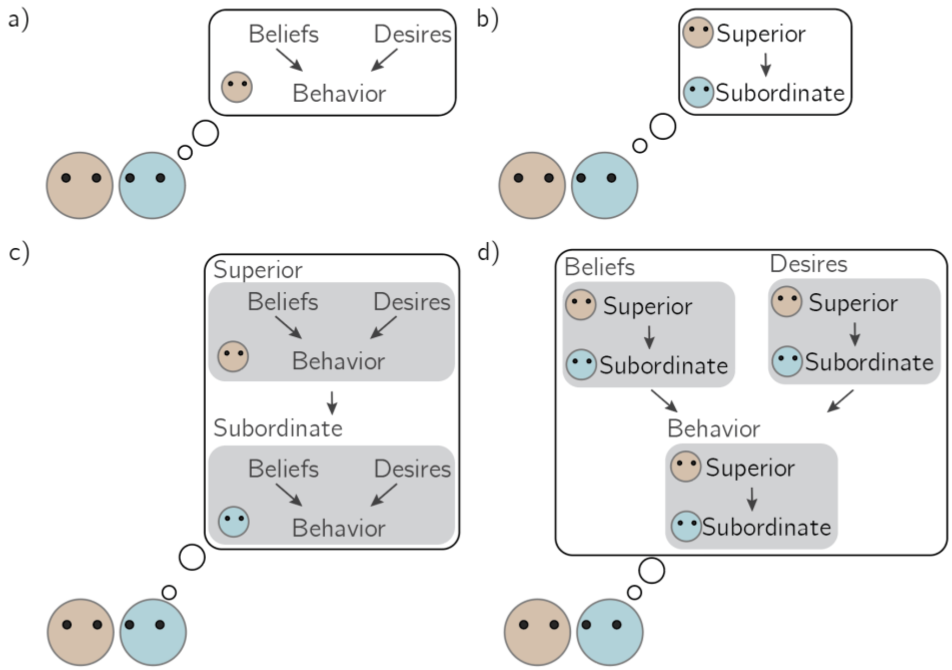

Figure 1. Schematic of an agent applying (a) a mentalistic stance (Section 2), (b) a simple institutional stance (Sections 3-4), (c) an institutional stance with mentalistic agents (Section 6.3), (d) a mentalistic stance to characterize and understand an institutional representation (Section 6.4).

There are four central aspects to our proposal. First, people routinely interpret and explain behavior by appealing to roles, norms, and scripts, and these representations cannot be adequately characterized as *mentalizing*. Second, these representations are not the byproduct of general intelligence, but instead reflect a fully coherent intuitive theory of social behavior. Third, representations within an institutional stance go beyond dyadic role representations. Instead, roles are interpreted in the context of a latent institutional representation that expresses how they fit into a broader system of social coordination (an institution). Fourth, the institutional stance is of comparable complexity to the mentalistic stance: It is evolutionarily ancient, emerges early in development, enjoys the support of specialized computations (and, perhaps, specialized neural circuitry), and it is uniquely human in important ways. We argue that, like the mentalistic stance, precursors of the institutional stance can be traced across multiple social species. The co-evolution of the *mentalistic stance* and the *institutional stance* can be thought of as providing complementary solutions to some of the same problems posed by social life. Mentalizing enables us to understand and predict each other's behavior across a wide range of contexts, in particular novel ones, based on a causal theory of how mental states relate to behavior. The institutional stance makes behavior intelligible and predictable within familiar contexts that are governed by shared norms of conduct, helping structure social behavior when mentalizing is costly or fallible.

While the institutional stance is grounded in non-mentalistic representations, this does not imply that it never interacts with a mentalistic stance. Quite the opposite. The power of the institutional stance in humans comes, in part, from our unique ability to combine it with complex mentalizing, enabling us to represent mentalistic agents acting within institutional structures, and to represent institutional structures as having mental states themselves (Figure 1). But as our opening examples illustrate, the two systems are co-equal in that neither can be fully reduced to the other.

Despite evidence of evolutionary precursors to the institutional stance across phylogeny, this representational system achieves its full power in humans. Non-human social species show the capacity to represent a fixed and predetermined set of institutional structures (such as kinship relations, rank hierarchies, and troop patrols). By contrast, the human institutional stance is generative, akin to a grammar (Crawford & Ostrom, 1995): We can represent an unbounded number of arbitrary institutional structures, from soccer teams, knitting groups and book clubs, to home-owner associations, universities and governments. In other words, the institutional stance in humans is a full-fledged and generative system.

The idea that institutions capture patterns of social behavior traces back to pioneers in sociology such as Émile Durkheim and Max Weber (Durkheim, 1911; Weber, 2004). Other work on social institutions (e.g. Boyer & Petersen, 2012; Teraji, 2018) has focused on large-scale constructs with explanatory power, or as cultural constructions that are built and refined over generations (corporations and governments, for example). More recently, cognitive scientists have begun to conceptualize roles and institutions more broadly as positions in social structures, and have explored how these structural phenomena can illuminate the causal structure of the underlying institutional concepts (Noyes et al., 2020a, 2021; Ritchie, 2020). From these important jumping off points, our project focuses on the intuitive cognitive representations underlying institutions, which function as a basic lens through which to understand and predict behavior (i.e., an interpretive *stance*). This is distinct from much past work, which instead seeks to characterize institutions from the perspective of social ontology (i.e., how we, as theorists, should understand institutions; Crawford & Ostrom, 1995; Guala, 2016; Searle, 2006; Hodgson, 2006). While that work is important and relevant for our concerns, it diverges from our project of characterizing the actual cognitive systems people use to understand each other's behavior.

In the remainder of the paper, we present the case for mentalizing as the foundation of human social intelligence (section 2), but argue that it fails to explain socially-complex behavior across species.

We propose that many forms of social coordination are enabled through role-based systems that form precursors to the institutional stance (section 3). We then formalize the institutional stance (section 4), and connect it to relevant frameworks (section 5). Finally, we discuss evidence for and implications of this framework (sections 6-7).

Table 1: *Glossary*
(A) Mentalistic stance glossary
Mentalistic Stance: An interpretative framework for understanding behavior in terms of unobservable mental states like beliefs, desires, and intentions. The mentalistic stance is often called an *intentional stance* (Dennett, 1989). Here we adopt new terminology for clarity, because the institutional stance also relies on representations of intentional (or at least teleological) action, with the difference that the intentions are thought to be generated by *roles* rather than by *mental states*.
Theory of Mind: The representational system used when taking a mentalistic stance. Theory of Mind consists of a mental causal model of how other people's unobservable mental states produce observable action, thought to be structured around a principle of rational action.
Teleological Stance: An interpretative framework for understanding behavior in terms of actions, goals, and physical constraints. This system allows observers to understand and predict simple behavior without having to represent complex mental states like beliefs and desires. Institutional representations can be built over a teleological stance, such that the institution represents roles in terms of goals, actions, and constraints.
(B) Institutional stance glossary
Institutional Stance: An interpretative framework for understanding behavior in terms of role-based institutional representations. In humans, the institutional stance is instantiated by a generative grammar of institutions which allows us to create and represent arbitrary institutional structures.
Institution: Central mental representation used to analyze and interpret how agents interact with one another. An institution is a network of roles and relations with normative entailments.
Proto-institutional representation: A hypothesized representational precursor to institutional representations. Proto-institutional representations are functionally equivalent to institutional representations, with the difference that the agent does not represent the distinction between roles and role occupiers and thus cannot represent the institution as a social construct.
Norm: Internal representational building block of institutional representations. A rule that grants, obligates, or restricts behavior, or that describes characteristic patterns of behavior within a role or across roles. Norms can be context-independent (applying to the role in all situations), context-dependent (becoming relevant only in certain situational contexts), and interaction-dependent (becoming relevant when interacting with specific other roles).
Role: Central representational unit in an institution. Agents occupy roles and then shape their behavior in accordance with its normative entailments and its relations to other roles. Roles are linked to other roles via inferentially rich normative relations, which together define institutions.

2. The case for mentalizing as the bedrock of social intelligence

Before presenting the institutional stance, it is helpful to briefly review the case for a mentalistic stance as a standalone cognitive system responsible for social intelligence. Taking a mentalistic stance is the activity of analyzing, representing, and interacting with social events via *Theory of Mind*—an intuitive theory positing unobservable mental states that cause behavior (Dennett, 1989; Gopnik & Meltzoff, 1997; Wellman, 2014). Evidence that the mentalistic stance is the bedrock of social cognition comes from four lines of research that broadly tell us: How does the mentalistic stance emerge and develop? What does mentalizing enable us to do that we couldn't otherwise? How is this capacity instantiated in the brain? And how ubiquitous is mentalizing in human social life? We briefly review answers to each of these questions below.

The first piece of evidence that mentalizing is central to social understanding comes from research showing that this system is functional from very early in life. Infants interpret other people's behavior in terms of unobservable mental states (Onishi & Baillargeon, 2005; Repacholi & Gopnik, 1997; Liszkowski et al., 2008) that rationally cause behavior (Gergely & Csibra, 2003; Liu et al., 2017; Woodward, 1998). This understanding structures early childhood cognition, including word learning (Akhtar et al., 1996; Vaish et al., 2011), language understanding (Ganea & Saylor, 2007; Jara-Ettinger et al., 2020; Saylor & Ganea, 2007), causal learning (Gergely et al., 2002; Bonawitz et al., 2011), socio-moral evaluations (Hamlin et al., 2013; Sloane et al., 2012), and epistemic vigilance (Harris, 2012; Aboody et al., 2022).

In adults, many complex social behaviors appear to rely on mentalizing. Formal models that instantiate basic forms of mentalizing capture with quantitative accuracy how people determine each other's preferences (Baker et al., 2017; Jern et al., 2017), predict behavior (Jara-Ettinger et al., 2020), cooperate (Wang et al., 2020), make social evaluations (Ullman et al., 2009), share knowledge (Shafto et al., 2014; Ho et al., 2016), and communicate (Goodman & Frank, 2016; Frank & Goodman, 2012).

Mentalizing is further supported by neural circuitry specialized for social computations (Saxe & Kanwisher, 2003; Saxe, 2006). These networks are directly linked to mental-state representations (Jamali et al., 2021); they are associated with social activities thought to require mentalizing, such as communication (Mi et al., 2021; Paunov et al., 2019) and moral reasoning (Young et al., 2010, 2007); and they correlate with computational models of human mentalizing (Collette et al., 2017).

Finally, some aspects of mentalizing may be ubiquitous and automatic. The basic building blocks—detecting agents, reading their intentions, and inferring their goals—are hardwired into human perception (Heider & Simmel, 1944; Gao et al., 2010; Colombatto et al., 2020; McMahon & Isik, 2023; Papeo et al., 2017), automatically triggering richer inferences about other people's mental states (Rubio-Fernández et al., 2019; Kampis & Southgate, 2020; Kovács et al., 2010).

3. Evolutionary origins of the institutional stance

While mentalizing is undoubtedly critical to human social intelligence, this capacity cannot be necessary for social behavior because complex social life is far more widespread across species than full-fledged mentalizing. Even in species with some form of Theory of Mind, such as non-human primates (Krupenye & Call, 2019; Royka & Santos, 2022), their mentalistic representations can be difficult to elicit, they are often brittle, and they lack the flexibility of human mentalizing (Martin & Santos, 2016; Horschler et al., 2023). How do social creatures that cannot build high-resolution predictive models of each other's minds coordinate and cooperate with one another?

The origins of the institutional stance reflect a pressure for living in social groups, combined with a limited capacity to understand or predict each others' behavior. Simply put, social species did not have the luxury of living in isolation until a sufficiently advanced form of mentalizing evolved (and indeed mentalizing itself might not emerge in full complexity in the absence of social life and its attendant pressures; Dunbar, 2009). What alternatives are available? The simplest solution is that two conspecifics solve social coordination challenges by developing a set of basic behavioral rules and regularities that make each other's behavior predictable and intelligible.

These representations still require some basic form of action understanding (at a minimum, a capacity to process body movements and make simple predictions such as direction of motion, but they could also be grounded on richer representations such as a *teleological stance*; Gergely & Csibra, 2003). However, these types of expectations do not require mental-state representations and can be expressed in terms of individuals occupying roles and engaging in rule-like behavioral patterns. Thus, a simple set of rules and roles allow conspecifics to understand and interact with each other without access to nuanced models of each other's mental life. Is such a view evolutionarily plausible? In what follows we argue that, across species, solutions to social coordination demonstrate that it is, and point to three critical stages for the emergence of a full-fledged institutional stance.

3.1 Stage 1: Proto-institutional representations lacking role-filler independent

The simplest type of representation that can support social coordination and interaction consists of *proto-institutional representations*: rigid role-based systems such as kinship relations and biologically-determined divisions of labor in which the assignment of individuals to roles is permanent (Figure 2a). In these systems, each member occupies a role that determines how to behave within a group and how to interact with others. These simple social structures can solve social interaction problems that would otherwise require mentalizing. For instance, a simple linear dominance hierarchy might encode a social affordance as basic as: in the presence of a conspecific, the higher-ranking member can do as they please while ignoring subordinates; subordinates must get out of the way and not interfere with the dominant conspecific. Simple rules like these are precursors to the institutional stance: role-based systems that support social interaction by determining what different agents can do within a social group, and how they ought to interact with one another. Such social structure would enable creatures to co-exist in a group without access to each other's minds as long as they can identify the roles they occupy and learn simple rules associated with them.

These rigid role systems appear as a solution across multiple social species, such as kinship relations, where conspecifics develop expectations about who will react and care for an infant in need (Dasser, 1988; Cheney & Seyfarth, 1980); primitive forms of in-group / out-group representations (Lorenz, 2021); or forms of coordinated hunting (such as with Aplomado falcons, where males and females perform different but fixed tasks; Hector, 1986). This type of proto-institutional system is common in social insects, where examples of role-based systems abound. One of the best known examples is the division of labor between workers and soldiers in ants (Hölldobler & Wilson, 1990), who act within biologically fixed social structures with clear roles that provide different social affordances. The rigidity of these social structures is particularly apparent in ants in the case of dulosis—the process where parasitic ant species capture worker ants from a different species, which then continue to mindlessly carry out their institutional roles now in the service of the parasitic colonizers (D'Ettorre & Heinze, 2001).

We call these *proto-institutional* because members of the social group do not, or at least need not, have any abstract representation of the social structure itself. That is, there is a lack of role-filler independence such that the role and role-occupier are not differentiated: associating different agents with different social affordances is enough (Figure 2a). At the same time, knowing what behaviors are associated with each agent still supports some forms of inference. For instance, the types of transitive inferences in a dominance hierarchy (e.g., those seen among pinyon jays; Paz-y-Miño et al., 2004) can be instantiated by simply reasoning about the relative power of different conspecifics. The cognitive prerequisites of a *proto-institutional* stance are also not trivial; an agent must know what social affordances are associated with their own role as well as with other agents. This can be achieved in a variety of ways, including individual identification (e.g., recognizing your offspring) or consistent visual markers (e.g., in European paper wasps—*Polistes dominula*; Tibbetts & Lindsay, 2008). It also requires knowing what behaviors are associated with roles, whether via biological hard-wiring or through real-time detection (e.g., Bumble bee workers 'reveal' dominance through fights; Amsalem et al., 2015. p. 40). This suggests diversity in how proto-institutional systems work, but the critical point is that these systems do not yet constitute an institutional stance.

3.2 Stage 2: Fixed institutional representations with role-filler independence

A more flexible form of role-based system—a *fixed institutional representation*—allows members of a group to swap roles (Figure 2b)[1]. This flexibility comes with increased representational and inferential demands via role-filler independence: Members must have an abstract representation of roles that is independent of the members occupying them; they must be able to infer which role a conspecific is occupying based on their appearance or behavior; they must track and remember role assignments and role changes throughout time; and they must know how to adjust their behavior as a function of the role that they themselves, as well as their interaction partners, occupy.

A candidate for such representation is male dominance hierarchies in monkeys, which are dynamic and continuously renegotiated. Indeed, some primates like Baboons are thought to have specialized intelligence for tracking and remembering rapid and complex role changes (Cheney & Seyfarth, 2019; Bergman et al., 2003). The variety of social behaviors non-human primates engage in also increases the demands for context-specific systems of organization, and it is possible that they have additional institutional representations such as troop patrols or even multi-role hunting structures (Boesch, 1994, 2005; although such interpretation is disputed; see, e.g., Moll & Tomasello, 2007).

[1] Here, we refer exclusively to role swapping happening *representationally* in conspecifics' minds. This is different from cases where conspecifics *physically* swap roles as in, e.g., physically changing one's sex in the case of sequential hermaphroditism (Gemmell, et al., 2019).

Another striking example comes from honey bees, which flexibly shift from foraging to receiving nectar. When returning foragers experience longer than normal delays finding a bee to receive nectar they often perform a tremble dance (distinct from the more famous waggle dance that indicates the source of nectar); observing foragers sometimes then switch roles to become receivers, maintaining an optimal balance between these two activities (Seeley, 1997).

These types of abstract institutional representations may be present beyond primates. In desert lions, for instance, female lions hunt through a coordinated structure that consists of two center hunters and two wings (Stander, 1992). While lions have role specialization in the form of favored roles that correlate with successful hunts, it appears that they can nonetheless switch between roles, suggesting a capacity to represent the different roles and an understanding that all roles must be filled for the hunt to succeed. Naked mole rats also show role specialization (Faulkes & Bennett, 2021), for example between defense, working to sustain the burrow, and caring for young. These roles sometimes shift as a function of age and body size but are also stable across individuals over time. Further, animals switch tasks to less preferred roles both when a preferred role is occupied (Gilbert et al., 2020) or when the animals previously fulfilling a role are removed from the colony (Mooney et al., 2015), usually in ways consistent with the deliberate selection of tasks that fulfill the needs of the colony.

While these representations are institutional—i.e., social interactions unfold via representations of interconnected roles within a set of agents—they do not yet constitute a full institutional stance because the set of institutional structures available to interpret behavior are pre-determined (Fig. 2c). There is no cultural evolution or emergence of novel institutional structures (di Fiore & Rendall, 1994). However, within these bounds, these representations are minimally generative in that the exact number of roles is mutable e.g., a dominance hierarchy expands and contracts based on group size. The advantages of institutional representations that can adapt to changing features such as population size may constitute an evolutionary pressure towards full-fledged institutional representations which can be more flexibly modified to meet a broader range of social needs.

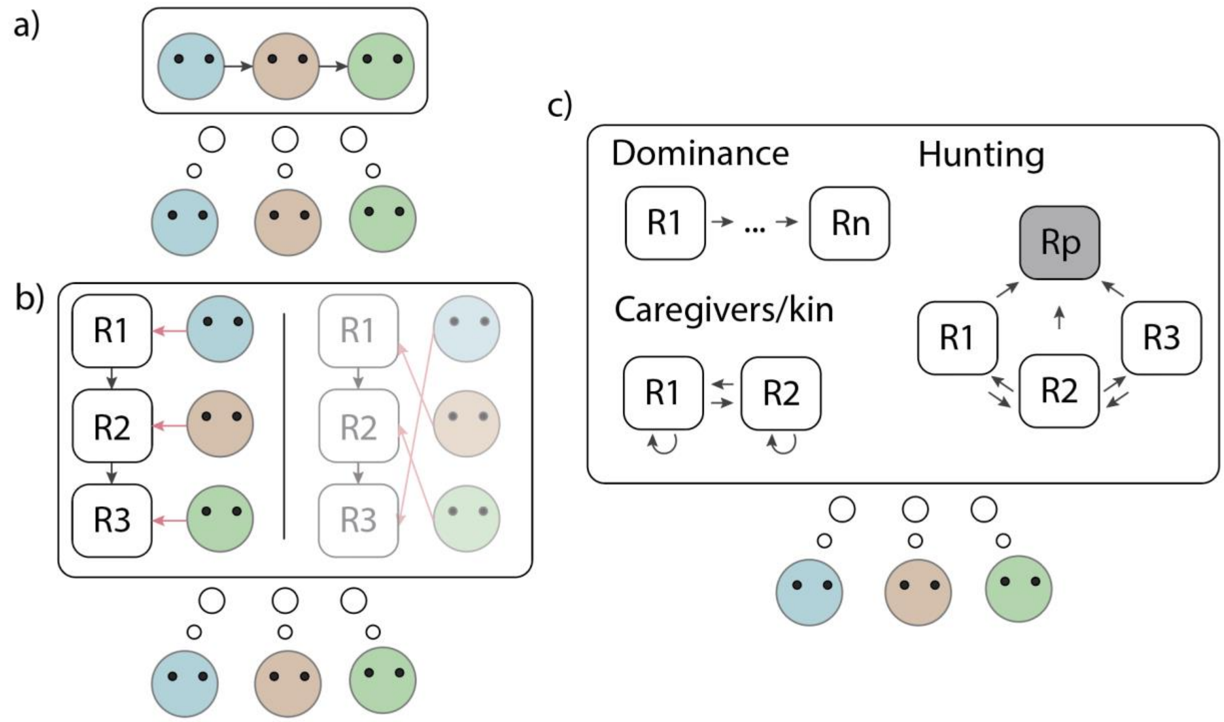

Figure 2. (a) Rigid proto-institutional structure where a group represents roles as intrinsically tied to agents. This type of implementation does not require users to have an abstract representation of the proto-institutional structure. (b) Institutional structure where a group has an abstract representation of the social structure, and a mapping from agents to roles. The ability to represent social roles as detached from individuals enables members to consider alternate arrangements and role switches. (c) Group with multiple institutional structures to structure a more complex social life. Here, the dominance representation is a linear structure that can be expanded or compressed based on group size; the caregiver/kin structure represents a system which includes expectations for how agents in the same role act towards one another (e.g. mutual expectations among siblings or among caregivers) as well as how agents in different roles act towards one another (e.g. expectations of care and deference). The hunting structure represents the prey (Rp) as a role in the system, despite the prey not being aware of this, as well as relationships between different hunter roles. As social complexity increases, conspecifics must represent a larger number of institutional representations, and consider how they might combine productively (e.g., how caregiver/kin relations affect conflict in or have critical entailments within dominance hierarchies). The increased number of institutional structures, and the need to reason about them compositionally, may have led to the generative institutional grammar in humans, which enables us to rapidly synthesize and infer institutional structures.

3.3 Stage 3: Generative institutional representations

This leads to the full-fledged institutional stance, which we propose is implemented uniquely in humans. The key difference in this final implementation is that the representations over institutions are fully generative: they no longer rely on a fixed set of evolutionarily ancient role-based systems. Instead, we have a generative representational system to synthesize novel institutional structures, leading to an explosion of role-based systems in human society, from reading clubs and sports teams to university committees and governments. This implies that we are equipped with a grammar for generating institutional structures, which can be built over complex representations of roles. In other words, we have an intuitive *Theory of Institutions*, akin to our *Theory of Mind*.

This is the institutional stance, which enables us to (1) identify normatively structured systems of interconnected roles, (2) compositionally construct new such systems, (3) determine what agents are currently occupying those roles, and (4) use that knowledge to interact with others. Indeed, the ethologists' ability to watch animal behavior and infer complex latent institutional representations at play is itself the institutional stance in use. Even if the cases we reviewed above turned out not to be precursor to the institutional stance, the propensity of human observers to interpret animal behavior in this way illustrates our tendency to read complex social behavior through latent representations of roles and role relations. We now formalize this proposal, review supporting evidence, and discuss its implications.

4. The institutional stance

We now turn to a formal description of the institutional stance. In brief, to take an institutional stance is to represent an individual or a group's behavior in terms of a rich set of interconnected norms that regulate, and guide how behavior unfolds over space and time. These norms are embedded in roles, i.e. nodes that agents can occupy. Roles are defined via both internal norms for how they operate and external norms defining how they relate to other roles. This network of interconnected roles constitutes the abstract representation of an institution. We describe each key component below.

4.1 What is an institution?

In its simplest form, an institution can be conceptualized as a graph where nodes represent roles and edges capture relations between roles (following Noyes et al., 2021; Ritchie, 2020a). Each node encodes norms internal to a role, including (1) who can occupy the role, (2) powers granted by the role, and (3) limitations or obligations imposed by it. For instance, a role might only be occupied by a particular sex (e.g., priests in the Catholic church), it might grant social powers (e.g. the ability of a member of a royal family to enter certain restricted spaces), and include obligations that are intrinsic to the role itself, independent of other roles (e.g., the role of a janitor to keep a building clean). Roles are then connected to one another via edges that specify the relational structure. These edges specify norms for how roles interact with one another. They can be diverse in content (e.g., reflecting power relations between a superior and subordinate, or relations of mutual support among coworkers); they can vary in directionality (directed as between superior and subordinate, or undirected as in coworkers at the same level), and they can grant social power over another role (e.g., a police officer's power to make an arrest over citizens).

An institutional representation is therefore a representation of the graph's *structure*, and to take on an institutional stance is to use knowledge of that structure as a causal model to interpret behavior. Critically, almost no roles are social isolates but rather are represented against the backdrop of a broader institution. This is true even if the full institutional structure is not represented. In the example above, when we represent an agent as acting in the role of janitor, we are aware that the role is situated within a broader system of roles. It is true that we often do not fully represent or know the details of this broader framework, but knowing of its presence is nonetheless crucial, helping us interact effectively with the agent and even to influence the system (e.g., if we wanted to change the time of day in which the janitor enters our lab we might speak with them directly but we might also consider whether the right intervention involves a discussion with someone else, say an office manager).

The institutional stance is an intuitive theory under the classical formulation of Gopnik & Meltzoff (1994): it supports prediction, interpretation, and explanation (i.e., the functional features of intuitive theories). The institutional stance also satisfies the four Gopnik-Meltzoff structural features of intuitive theories. First, representations of institutions are *abstract*: the representations used to interpret behavior (e.g. roles and norms) are different from the representations of the data being explained (observable action). Second, these representations are *coherent*: the theoretical constructs do not work independently, but instead operate together as a single system, where multiple components are needed to interpret behavior. Third, the representations are *causal*. That is, an institutional stance represents roles as having *causal* and *explanatory* power in the counterfactual sense (i.e., we expect the behaviors would not have occurred had the agent not been occupying the role, and explain it accordingly; Pearl, 2009). Finally, the representations carry *ontological commitments*. That is, applying an institutional stance comes with a commitment to its predictions, and we use the system to make counterfactual claims.

Note that ontological commitment in an institutional stance reflects a belief that the structure is capturing real social-causal structures. But this does not require that the agents represent themselves in the same way. For instance, a person ordering coffee might take an institutional stance to interpret their barista's behavior, and this stance would in some sense be real: the barista is in fact acting under a role, and their behavior reflects that. However, the barista might also find it useful to take an institutional stance towards the customer, even though the customer might not be thinking of themselves as playing a particular role. Nonetheless, this is possible because the systematicity in the goals and behaviors that customers have makes them easy to represent through a role, providing a set of expectations that do not rest on an analysis of each individual's mental states.

4.2 Institutional representations are not unstructured collections of norms

What is the internal representational content of roles and edges through which they guide social interactions? In our proposal, the basic building block of an institutional stance is a representation of norms, sorted based on whether they apply to an individual in a role (node norms) or to a relation between individuals in roles (edge norms). It is thus worth considering whether our proposal can be reduced to a simple observation that people know that norms affect behavior and adjust their inferences accordingly.

By most prominent accounts, complex normative systems are critical for explaining the parallel emergence of elaborate and stable systems of cooperation. Norms benefit cooperation in three interrelated ways. First, shared norms reduce coordination costs by making behavior predictable; if I know what you are likely to do I can easily adapt my behavior to be complementary (and you can do the same if you know what I am likely to do). Second, because norms facilitate cooperation in this way, failing to adhere to a norm can undermine cooperation and affect others. Routine norm violators may therefore face reputational costs or other forms of retaliation, and knowledge of this further incentivizes compliance. Third, given potential costs associated with negotiating ad-hoc norms on the fly, norms tend to stabilize within groups, allowing coordination to proceed more easily.

In classical literature, norms are broadly thought of as falling into two categories: descriptive and injunctive (Kallgren, et al., 2000; Lapinski & Rimal, 2005). Injunctive norms are characterized as being subject to social pressure and risk of punishment or retaliation when violated (e.g., being quiet at a library). Descriptive norms represent stable emergent statistical patterns of how people behave without enforcement (e.g., clapping at the end of a theater play). The institutional stance invokes a type of *institutional norm* that does not cleanly map onto the descriptive/injunctive distinction, in that it has three key properties. First, institutional norms are role-dependent, such that their scope is defined by the role one occupies within an underlying institutional structure. Second, institutional norms are interpreted as causal: we see norm adherence as being causally generated by a latent role in line with intuitive counterfactual causal models (i.e., the belief that if they were not in the role they would not have behaved in that particular way). Third, collections of norms within a role are coherent. That is, roles are not an unsorted and random assortment of norms. Instead, the collection of norms embedded in a role form a coherent set that is in the service of supporting fluent social interactions within the context of a full institutional structure.

Many of these norms may well be special cases of descriptive or injunctive norms. But our point here is that reasoning about when a norm becomes relevant or active in social settings requires a latent representation of the institutional structure; that is, to know what institutional norm applies you need to know what institutional structure you are within. It is also the case that reasoning about norms within an institutional stance has important additional entailments that follow from institutional knowledge, and which are not generally present for stand-alone injunctive norms. For example, when someone violates an expected institutional norm it may indicate that the agent has left the role or that they have incomplete knowledge about it, rather than implying that their behavior should be punished.

4.3 Intuitive institutions, theories of institutions, and real institutions: What's the connection?

What is the relation between the institutional stance, scientific theories of institutions, and the real-world structure of legal institutions? To begin, the building blocks of our framework—roles and norms—are similar to those used in formal theories of institutions developed in social sciences and philosophy (e.g., Crawford & Ostrom, 1995; Hodgson, 2006). We believe this is no coincidence. In the beginning of any formal endeavor, theoreticians' have no other way to evaluate the plausibility of a proposal than by comparing it against their intuitions. Inevitably then, seminal formalizations of different phenomena often share crucial similarities to our intuitive theories. To give just two examples, human intuitive physics is surprisingly similar to the first formalizations of physical laws (Battaglia et al., 2013) and, within mentalizing, intuitions about how others make decisions are also strikingly similar to principles of classical economics (Jara-Ettinger et al., 2016). The origins of a formal inquiry of social ontology may be no exception. If correct, looking into how early theoreticians formalize institutions could hold valuable clues as to the nature of our intuitive theories of institutions (e.g., von Pufendorf, 1759; Hobbes, 2016).

The relation between the institutional stance and real-world institutions might also run deep. The ability to *create* new institutions gives us the ability to decide to act according to them, and make them *real* (a point that we expand on in Section 6.4). At the same time, once an institution is created, the institutions themselves are subject to cultural evolution and mutation, and can be modified by many agents in parallel, exploding into systems that are too complex for any individual member to represent or understand. This supports the emergence of real-world institutions that exceed the ability of any one agent to fully comprehend but that remain locally understandable via an institutional stance (for example academics may have a good grasp of the institutional structure of their department but find the structure of the university far too massive to fully comprehend). This highlights the distinction between our focus on the institutional stance as an intuitive theory of person perception and the analysis of formal institutions that other social scientists might undertake.

5. Relation to other theoretical frameworks

The proposal that we have presented shares important similarities—and differences—with related frameworks that have been proposed in social cognition and which we briefly review here.

5.1 Schemas and scripts

Schema and *script* approaches to social cognition (Bartlett, 1932; Eickers, 2023, *in press;* Mandler, 2014; Taylor & Crocker, 2022) constitute a venerable class of theories with a close connection to our perspective. Schemas refer to knowledge that structures and organizes a particular behavioral domain, which is sometimes analogized to a dramatic script. For example, a supermarket script might describe the types and sequences of activities that take place in a supermarket. This knowledge is organized via the roles that frequently occur in that setting as well as the activities that characterize those roles. For instance, it might specify the role of *shopper* in terms of activities like getting a shopping cart, gathering groceries, going to the registrar, paying, and leaving with groceries; a role of *cashier* in terms of things like scanning each item, placing items in bags, accepting payment, and so on.

The institutional stance converges with this framework in the claim that a powerful way to organize social information is via normatively defined roles and their relations. The central claim of the institutional stance goes beyond this, however, in arguing that this is not merely a pragmatically useful way of organizing social information. Rather, it reflects an evolutionarily ancient specialized intuitive theory for the rapid acquisition and deployment of social structures, i.e., institutional representations. By contrast, script approaches are considerably broader, providing a domain general account of how information governing familiar domains of behavior is organized. One can in this sense have a script for solitary and non-social activities such as building a campfire. In our framework these non-social scripts should be differentiated from those described via the institutional stance, even if they may well share common structure. Both representations might be the result of convergent cognitive evolution. Non-social scripts might alternatively be an outgrowth or a side-effect of the institutional stance. Or, conversely, perhaps non-social scripts emerge first, and form a precursor to the types of scripts needed to represent the types of social scripts that build the first stages of proto-institutional representations.

5.2 Folk Sociology

A second related approach to social cognition has argued that humans have a *folk sociology*— a framework for interpreting behavior in terms of social relations, with a particular emphasis on groups (Rhodes & Baron, 2019; Hirschfeld, 2013; Clément et al., 2011; Shutts & Kalish, 2021). Work in folk sociology has typically emphasized essentialized groups that are stable and immutable—i.e., thought to be determined by patterns of deep internal, perhaps genetic or biological, similarity (Gelman, 2004; Rhodes & Mandalaywala, 2017), most prominently race and gender (e.g. Prentice & Miller, 2007). These group representations help us identify social types, evaluate them, and develop shared expectations for how group members behave (Shutts & Kalish, 2021). Importantly, recent work has shown that these phenomena are not restricted to essentialized social categories. Both children and adults believe that patterns of existing social agreement (Noyes et al., 2017), as well as formal membership criteria (Noyes et al., 2020a), can underlie group membership. What's more, they assert that social and cultural forces can produce highly meaningful and inductively rich social kinds (Noyes & Keil, 2020b).

Another critical aspect of folk sociology is the tendency for even young children to divide the social world into categories representing "us" versus "them", and positively evaluate the "us" in myriad ways (Dunham, 2018). This type of reasoning about social groups has of course also been a central topic in social psychology in the study of stereotyping, prejudice, and discrimination. At first sight, these classical instances of stereotyping and prejudice appear to fall outside the scope of the institutional stance. However, some philosophical accounts argue that even the most frequently essentialized social categories such as race and gender are better conceived of in structural terms, i.e. as systematic patterns of hierarchical subordination such that to be conceived of as a woman or racial minority *is* to be relegated to subordinate status (Ásta, 2018; Haslanger, 1995, 2000). On such accounts essentialism is not merely explanatory but also ideological: By positing deep-seated biological differences it seeks to render unimpeachable and natural differences that are actually structural differences caused by power and status differentials. An institutional approach to these categories thus becomes an important theoretical lens for ameliorative projects.

Some work has also emphasized that many social groups are better defined via deontic or normative properties than internal essences (i.e. via rights and obligations assigned to members; e.g. Foster-Hanson & Rhodes, 2019; Kalish & Lawson, 2008). However, this perspective generally treats social groups as undifferentiated containers of group members in which exchangeable individuals inherit properties (whether traits and behaviors or rights and obligations) directly from the group's attributes. By contrast, the institutional stance posits *structured* social systems, i.e., how individual roles are related to other roles in specific ways (see also Noyes et al., 2021). Structure of this sort frequently exists *within* groups as well (consider the complicated normative entailments we can derive from the structured relationships between junior and senior members of the same academic department, for example). Thus, the institutional stance provides a finer grained account of within-group differentiation, while also providing the tools to understand structured relationships between groups, such as those existing within a vertical status hierarchy. Under this view, these broad group representations might also reflect broad instantiations of the institutional stance where groups are defined by the role of a group occupier. It also dovetails with more recent work taking seriously the structured relationships between groups and the distinct patterns of reasoning that emerge from them (Amemiya et al., 2022; Vasilyeva & Lombrozo, 2020 ; Vasilyeva et al., 2018; Yang et al., 2021). These examples support a powerful human ability to reason about groups as positions in structured institutional wholes (Noyes et al., 2021; Ritchie, 2020a), which has not received much attention in the folk sociology literature.

5.3 Shared intentionality

A third related framework comes from work on the development of normativity and cooperation, which has argued that complex social coordination is instantiated by a capacity to share intentions (in psychology: Tomasello, et al., 2005; Tomasello & Carpenter, 2007; in philosophy: Searle, 2006; Gilbert, 1992). The key idea is that cooperative behavior often consists of individual goals which together fulfill a superordinate goal. According to this framework, fulfilling the superordinate goal requires members of the cooperative interaction to represent a shared intention that they are fulfilling together. Consider the task of collaboratively opening a cardboard box, in which one person holds the box while the other cuts it open (Tomasello et al., 2005). Each agent has a set of focal goals they must keep in mind (hold the box steady; cut the tape with the box cutter) but they must also keep in mind the goal of their partner, and adjust their behavior to comply with the higher and lower-level implementation their partner adopts. Once this entire structure is represented, role switching is unproblematic; the holder can become the cutter and vice versa.

Proposers of shared intentionality have argued that this capacity is unique to humans (Tomasello & Carpenter, 2007; Tomasello & Herrmann, 2010; Tomasello & Moll, 2010) and that it is a cognitive prerequisite for any type of role-governed cooperative behavior (Moll & Tomasello, 2007; Tomasello, 2022). Therefore, this view might imply that the institutional stance is unique to humans, thoroughly mentalistic, and therefore unable to support non-human social behavior. But this is not necessarily the case. Shared intentions are undoubtedly critical to the ad-hoc creation of much role-based behavior, enabling people to rapidly invent new roles and institutional structures, supporting the unique generative capacity of the institutional stance in humans. At the same time, some institutional representations (like those reviewed in Sections 3.1 and 3.2), might be biologically determined, such that conspecifics can understand each others' behavior in terms of role-based structures without necessarily thinking of these roles as having a goal or understanding how they fit together. In this sense, the roles within the institutional stance are not *Tomasellian roles* (Tomasello, 2020), although they can also include shared intentions. Instead, these representations are consistent with the notion of *minimal cooperation* (Ritchie, 2020b), in which individuals playing roles in organized structures can, by simply playing their role, achieve higher order forms of coordination as long as others also play *their* roles. Of course, this only works in familiar contexts, in that it requires knowledge of the roles and their corresponding behavior (or at least one's own role, which would be sufficient as long as everyone knows and enacts their own role such that everything goes to plan). But it transforms a case of collective intentionality or recursive mental state modeling into one of rule following, i.e. of norm-adherence within a role.

6. Institutional stance in use

Having defined the institutional stance, we now turn our discussion to questions that arise about this framework: How does the institutional stance emerge and develop (Section 6.1)? How is the institutional stance deployed in everyday life (Section 6.2)? How do the institutional stance and the mentalistic stance interact (Section 6.3)? And what kinds of social behaviors does the institutional stance uniquely explain (Section 6.4)?

6.1 How are institutional representations acquired?

Given that simple role-based social reasoning is evolutionarily ancient (section 3), we expect even human infants to have the basic role-based representational primitives that support the institutional stance. But humans must be further equipped with the means for learning new institutional representations that support cultural innovation and the ability to immerse ourselves into complex social systems that vary across societies. The current experimental record provides support for this view.

First, role-based social primitives are at work quite early in development. Infants have basic and stable dominance representations that guide their social predictions (Enright et al., 2017; Mascaro & Csibra, 2012; Thomsen, et al., 2011; Stavans & Baillargeon, 2019), they distinguish dominance coerced through fear from dominance that's been socially granted (i.e., role-like; Margoni, et al., 2018; Thomas, et al., 2018), and make transitive inferences from pairwise dominance displays (Gazes et al., 2017). Another domain that points to early role-based reasoning comes from research on caring relations—which are difficult to understand under standard mentalistic frameworks (Gopnik, 2023). Infants understand caring relations (Spokes & Spelke, 2017), but their predictions about caregivers are shaped by their own experience; infants that are *securely attached* (Ainsworth, 1979) have stronger expectations that a caregiver will comfort a distressed agent (Johnson et al., 2007, 2010). This suggests both an early capacity to represent evolutionarily-ancient roles and some ability to flexibly adjust expectations about how exactly they operate.

This research reveals that infants already represent content-rich roles that are present across species, but additional research suggests they might also be able to represent more abstract forms of roles (i.e., detecting roles that are not necessarily tied to evolutionarily ancient ones like dominance or caring). Indeed, infants encode simple social interactions in the context of latent social structures (Powell & Spelke, 2013; Thomas et al., 2022), and find it surprising when an agent and patient swap behaviors in an interaction, suggesting that they interpret it as a role reversal (Golinkoff & Kerr, 1978; Schöppner, et al., 2006; Tatone, et al., 2015). Moreover, infants seem overly sensitive to signals that might reveal social roles. When watching dyadic interactions, surprisingly minimal cues of object transfer trigger a representation of an actor *giving* an object to a patient (e.g., an object being displaced towards another agent as a side effect; Tatone, et al.,, 2019), and such representations lead to improved object encoding (Tatone,et al.,, 2021). Most surprisingly, infants seem to be readily attending to the existence of latent social categories—evidenced by Tseltal infants in southern Mexico learning their languages' honorific terms from indirect passive exposure (Foushee & Srinivasan, 2023).

When infants transition to toddlers and become active members of the social world, they spontaneously generate basic social structures (such as hierarchies similar to the ones seen in non-human primates; McGrew, 1972; Sluckin & Smith, 1977), and they already make different social predictions based on whether mental states or norms are play. For example, in a series of studies (Clément et al., 2011), 3-yr-old children show the classic failure to reason about false beliefs by predicting that a person will look for an object where it actually is rather than where, based on the available information, the person ought to believe it is. But when this same dilemma is framed as the application of a rule (e.g. dolls are always in the toy box) these same 3-yr-olds expect the agent to look in the toy box even though the children themselves know it is no longer there.

At this age, children are also cognitively prepared to learn institutional representations. Young children have better knowledge and memory of role-based events than specific events (Fivush, 1984), and their ability to make inferences from these schemas (such as whether the shoppers paid the cashier) surpasses their ability to make simple logical inferences or inferences from general world knowledge (Hudson & Slackman, 1990). Moreover, much of children's pretend play focuses on enacting *sociodramatic role play*—pretend play using real-life roles such as caregiver, shopkeeper, doctor, or schoolteacher, as opposed to fictional ones (Christie, 1982; Corsaro, 1993; 1979), and children engage in these activities in part because they are unable to actually do them (Taggart, et al., 2018; 2020). This sociodramatic role play might be a form of early deployment and practice of the institutional stance. Indeed, while children are often characterized in terms of their rich imaginations, it has been noted that much of what they actually imagine is relatively mundane, frequently focusing on the enactment of role scripts like the ones we've described here (Harris, 2021).

Children are also strongly motivated to learn and enforce social norms (Partington et al., 2023; Schmidt et al., 2016; Rakoczy et al., 2008, 2013; Schmidt & Rakoczy, 2023). While this capacity has usually been studied in the context of isolated norms, this same capacity might underlie the ability to also quickly detect and learn norms within institutional representations. Indeed, children generally think of norms as characterizing a particular group rather than all individuals (Schmidt et al., 2012), and can rationally infer a norm's scope (i.e., which group or role it applies to) by considering the statistical likelihood of seeing norm-consistent behavior in different agents (Partington et al., 2023). While this work has focused on group scope, similar mechanisms might allow observers to detect role-related norms.

Learning institutional representations might be further boosted by people's general ability to detect the covariation between behaviors and situational contexts (Kelley, 1980; Seiver et al., 2013; Ruffman et al., 2012). Here again, this work has largely explored how people infer the extent to which people's behavior might be the cause of a situational context (rather than an intrinsic and stable mental state), but these same principles might underlie the ability to learn a role. For instance, imagine walking into a building and having a person greet you. You might initially assume they were just being polite, but if you continue visiting the building throughout the year, and notice that there is always a person there waiting to greet you, this pattern of covariation reveals the presence of a role.

Furthermore, the ability to mentalize likely makes learning institutional structures easier, particularly for adults. Our mentalistic stance includes general expectations about what types of mental states agents generally have. Consider a doorman: if the person simply greets you and does nothing else, you might initially not realize they were playing a role, because a generally friendly attitude towards others is not an unusual mental state. Imagine instead that you saw the person rush to the door as soon as they spotted you, open it for you (despite the door being clearly unlocked), and hold it open as you walk in. While this behavior can be interpreted mentalistically, it might feel unlikely that a stranger would have such a strong motivation to open the door for you, particularly if it's in a case where you could have done so easily yourself. This low likelihood can thereby serve as a powerful signal to the presence of a role being performed, in turn telling us that we ourselves are being treated under an institutional stance as well—e.g. as a guest or customer—and affecting how we react (such as by explaining our purpose in the building; something that we wouldn't volunteer to a stranger passing on the street). While these strategies might be particularly powerful for adults, they might also be available to children in some capacity. For instance, young children have expectations about what types of costs are reasonable to incur for others (e.g., ask for help only when the helper can accomplish the goal more easily than you could alone; Grosse et al., 2010; Jara-Ettinger et al., 2020). Violations of such expectations might help reveal roles and the tasks they perform.

Finally, recent work suggests that adults can perform Bayesian inference, or some approximation thereof, to rapidly infer latent social structures, such as group boundaries (Lau et al., 2018; Gershman & Cikara, 2020). What's more, given a known set of roles (e.g., managers and employees), people can reconstruct the exact type of institutional structure (e.g., exactly how many managers there are, and who they have oversight over), as well as which agent is occupying each role, through relatively sparse observations of agents interacting with one another (Davis, et al., 2022). All of this work, together, might point to specialized reasoning systems that humans have specifically for reading institutional structures (akin, to the apparent specialized inferential and neural circuitry for reading mental states; Saxe & Kanwisher, 2003).

6.2 When do we deploy an institutional stance?

Granting the presence of an institutional stance also raises the question of how and when we deploy it, especially given our clear ability to analyze behavior in terms of mental states. Indeed, determining when to deploy an institutional or mentalistic stance, or a combination thereof, is challenging in principle because many behaviors can be explained *either* as the expression of a role *or* as a combination of an appropriate belief and desire.

At one extreme, it is possible that the institutional stance is pervasively and automatically operating in all or nearly all social settings. This would follow if its evolutionary history and centrality to social cognition runs as deep as mentalizing, which has often been described as automatically deployed (Kampis & Southgate, 2020). Even in the cases where we are richly mentalizing about others, latent scripts might nonetheless structure some form of our interaction—basic role-based representation such as 'friend', 'co-worker', 'family member', or even 'stranger' provide a set of expectations that could greatly simplify social interactions (e.g., they might be a source of trait attributions that help us constrain and simplify mentalizing; Westra, 2018, 2020). Related ideas have been elaborated on in the social sciences, for example via Relational Frames Theory, in which a set of prototypical relationship types pervasively and even automatically affect social cognition (Fiske, 1992). With this in mind, return to our paper's opening example, where Kate asks Eric to fulfill certain roles when they run into each other at the cinema. The surprise of the untowards request could come not only from knowing that the previous roles are no longer operative, but also because we know and are automatically tracking new roles— 'acquaintance' or 'friend'—that have very different entailments.

Even if some form of the institutional stance is pervasively active, some roles carry tighter social scripts and have higher stakes than others (e.g., *police officer* vs. *friend*). This creates a motivation to ensure that our own behavior is appropriately interpreted through an institutional stance when relevant, and in many cases we broadcast this information to help resolve any potential ambiguity.

Figure 3. Two identical agents taking identical actions in the same context. The single presence of an apron disclosing an institutional stance triggers different inferences, predictions, and expectations.

Indeed, individuals (and institutions) often take explicit steps to signal the framework within which they want to be interpreted, via “status indicators” (Searle, 2006). Take a guard in charge of preventing access to a building; they likely want observers to read their behavior through an institutional lens, rather than to attribute incorrect and potentially adversarial mental states to them. Visible markers like uniforms—which are pervasive in the social world, such as those worn by police officers, priests, lifeguards, doctors, and baristas—are one notable way we do this. Consider Figure 3, which shows two identical agents taking identical actions in identical contexts. The simple presence of an apron leads to a radical revision to how we interpret the behavior and mental states of the agent on the right. In the first case, we might infer that the person likes olive oil, intends to get some, and is likely to cook a meal with it that week. In the second case, the same behavior no longer reveals any personal attitude towards olive oil, and we instead read the behavior as revealing a role-based task, such as ensuring that the oil has not yet passed the sell-by date.

In related cases agents signal their institutional status in other ways, most notably verbally. Emails may begin with a disclosure such as “I’m writing in my capacity as Chair of the Search Committee” to indicate that the action one is taking (perhaps rejecting a candidate) should be attributed to the carrying out of an institutional role rather than the agent’s personal beliefs and desires. In some such cases speakers even explicitly disavow the mental states that might normally be inferred from the behavior (“it gives me no pleasure to have to write this letter…”), a form of speech that can also be interpreted as urging the recipient to interpret their behavior as the enactment of a role demand rather than a personal desire. These institutional disclosures are even deployed in more informal roles: consider someone in a heated debate declaring that they are simply “playing the devil’s advocate”.

Beyond physical institutional indicators (e.g., uniforms) and linguistic constructions (e.g., explicitly disclosing an institutional role), markers of an institutional stance might sometimes even be embedded in language itself. For instance, many languages use different pronouns to encode some type of formality (Brown & Gilman, 1960; Goffman & Giglioli, 1972) in which more formality tends to coincide with more institutional roles. Examples of this include the *T-V distinction* in Indo-European languages (e.g., *tú/usted* in Spanish), honorifics in Japanese in which relative status in encoded in verb endings (McCready, 2019), or Nahuatl honorifics (a four-level morphological system; Hill & Hill, 1978). It is possible that these linguistic markers are used not only for deferential purposes but to signal when an engagement ought to be read through an institutional stance rather than a mentalistic one. Another example of language as a signal to role comes from surnames: many surnames historically emerged to indicate occupations (e.g., *smith*, *abbot*, *baker*, *butler*, *carpenter*, *cook*; Gersuny, 1974), and this might have been precisely to help disclose institutional roles.

However, there are also cases in which people must determine whether to adopt an institutional stance in the absence of clear status indicators. In these cases, knowledge of the institutional structure that one is operating within likely helps us infer the relevant roles. Imagine, for instance, walking into a coffee shop and having a person in plain clothes greet you and ask you what you want. In this case, the low likelihood that a stranger would be motivated to spontaneously inquire about your general desires, combined with our knowledge of the roles typically found in coffee shops and our awareness that they are frequently not visibly marked, is enough for us to identify a role agent and so adopt an institutional stance with respect to them. In cases like these, it may be the fact that the institutional structure is assumed to be well known that leads to a decreased use of institutional markers.

Of course the strategies described above, both identifying institutional indicators and inferring roles from knowledge of institutional settings even in the absence of those indicators, only work for enculturated adults with rich knowledge of the institutions they encounter. It cannot account for actions within highly unfamiliar or entirely novel institutional structures (a domain which, even if modest in adulthood, will be vast earlier in development). This makes the mechanisms for inferring the hidden institutional structure reviewed in 6.1 critical, a process likely boosted by explicit cultural transmission concerning institutions and their central roles.

6.3 Interplay with mentalizing

6.3.1 Institutional stance without mentalizing

Given that humans have a mentalistic stance and an institutional stance, how do the two interact? To start, in the same way that the institutional stance emerged in other species as a solution to coordination in the absence of Theory of Mind, the institutional stance in humans can reduce the cognitive costs and uncertainty associated with mentalizing: Structured knowledge creates the possibility that we can effortlessly engage in an otherwise complex interaction without invoking representations of beliefs or desires. For instance, if you order a coffee at a coffee shop and interpret the barista under an institutional stance, you might then simply assume that the goal will be fulfilled, and that a coffee will materialize within a few minutes, without the need to actively track and infer the barista's mental states to determine whether the barista indeed has the goal of preparing the coffee for you or not. This could be broadly similar to how we can interact with a vending machine expecting an outcome, such as (an admittedly awful) coffee to appear, without attributing a rich mental life to the vending machine. In both cases, this is possible because of a structured understanding of particular contexts defined at least in part via how our actions in one role will tend to produce reactions in other roles.

There are a few lines of research suggesting that an institutional stance could effectively reduce the burden of mentalizing. Indeed, the development of scripts as a tool for understanding social cognition came from early practitioners of artificial intelligence (Schank & Abelson, 1977) looking for a means of creating socially interactive agents that explicitly lacked true mentalizing abilities. And game theoretic contexts that traditionally required mental state coordination (who will lead or follow the dance, and do we all share that common knowledge?) can be transformed into non-mentalistic cases of rule following by conditioning behavior on a rule invoked via some feature of the context initially external to the game (e.g., men lead and women follow) (Guala, 2016; O'Connor, 2019). Further, when an observer knows the structure of possible actions an agent might take in a situation, action understanding becomes computationally simple, no longer requiring complex Theory of Mind (Davis & Jara-Ettinger, 2022). Clinical research also suggests that, people with Autism Spectrum Disorder—often characterized by difficulties in making use of mental state information (Baron-Cohen, 2000)—benefit from being taught detailed script knowledge for a range of social interactions (reviewed in Akers et al., 2106), again suggesting that the institutional stance is distinct from mental state reasoning, which is is often difficult for individuals with ASD. Finally, some empirical work does suggest that norm-based systems for predicting behavior need not rest on mental state reasoning (Kaufmann & Clément, 2014). Together, this work suggests that role-based reasoning within the institutional stance does not directly require mental state reasoning, and can be handled without deploying a mentalistic stance at all (although, naturally, the social representations provided by perception—notions of agent, goal, and intention—ought to be readily available to the institutional stance; McMahon & Isik, 2023).

6.3.2 Institutional and mentalistic stance in competition

Because many behaviors can be explained either through a mentalistic or an institutional stance, it is natural that these two frameworks might often compete. In these cases, an explanatory or predictive failure of one system can trigger inferences in the other. That is, if an institutional prediction fails we ascribe mental states, and if a mentalistic explanation fails we are more likely to ascribe institutional roles. Returning to the coffee shop example, suppose now that you've been waiting for ten minutes and your coffee is still not ready. You might then start attending to the barista to try to figure out what's going on. If you see them working hectically you might infer that they just haven't gotten to your order yet, whereas if you see them relaxedly talking to a coworker, you might infer that they either lost or forgot your order. In other words, deviations from role-based expectations (like deviations from other normative expectations) likely trigger a shift towards greater mental state inference, deployed in an effort to make sense of the deviation. Indeed, several analyses of mindreading have posited that its use might become active precisely when non-mentalistic expectations fail (Bohl, 2015; Zawidzki, 2013).

Conversely—although perhaps less commonly—violations of a mentalistic expectation might trigger institutional expectations. Imagine, for instance, that you are friends with the barista, but notice that they are acting unusually formal towards you. Such a deviation might trigger us to consider whether their behavior is meant to be interpreted under the institutional stance. These switches can happen frequently within a single interaction. If the barista were your friend, you might quickly alternate between catching up and placing an order, which would require rapidly knowing which aspects of the interaction are institutional and which are mentalistic. Our ability to do this suggests that we are able to have active parallel representations of agents' mental lives and their institutional roles.

6.3.3 Combining institutional and mentalistic representations

Finally, the ability to represent other people's mental states enables us to create and represent richer types of institutional structures than would be otherwise possible. This is because we can encode expectations about mental states given relative roles (Fig. 1c). For instance, we can express interactions where one agent's role is to satisfy another agent's general desires (such as in the office example in the introduction), or where occupying a certain role requires having certain mental states (such as expecting a stock person at a supermarket to know where items are located). In many of these cases, even if the roles require us to represent mental states, these are tagged accordingly. For instance, in the supermarket stock person example, we might locally represent them as *wanting* to stock the shelves quickly to complete their shift early, but we are simultaneously aware this is not because they find this activity intrinsically rewarding, rather because they'd much rather be doing something else. Similarly, their detailed knowledge of where things are stocked would not lead us to infer that they must be passionate about supermarkets—an inference that might be warranted if this knowledge were acquired in their free time. At the same time, we suspect there is widespread variability in how readily we transfer behaviors or goals from a role to an agent. For example, in baseball, the pitcher's goal is to successfully throw the ball to the catcher without the batter getting the chance to hit the ball. In this case, our institutional representation of the role of pitcher does seem to intuitively transfer a goal of striking out a batter to the agent in that role. That is, the pitcher is much more likely to truly want to accomplish their roles' goals than is the stock person (Nguyen, 2020).

A second way in which mentalizing and the institutional stance combine is through the attribution of mental states to institutional representations (Fig. 1d). The propensity to ascribe mental states to groups—institutional or not—is common (e.g. Russia *wants* to annex Ukraine, and Ukraine *intends* to resist; Jenkins, et al., 2014; Tang & Gray, 2023), and this might reflect a general capacity to anthropomorphize groups and ascribe mental states.

6.4 Social Constitution

So far we have argued that the human institutional stance is unique in that it is a generative intuitive theory, enabling us to synthesize new institutional structures rather than being limited to a small set of predetermined ones. This implies that we can explicitly represent institutional structures and reason about them, rather than simply reasoning *through* them (as one would through a proto-institutional representation like the ones reviewed in Section 3). The ability to represent institutional structures as objects of thought might be a key precursor that helps explain two uniquely-human aspects of social intelligence.

First, the capacity to read complex multi-agent behavior through an institutional stance means that we can apply this capacity as an analytic tool, including to analyze phenomena that we might not initially think of as being institutional—i.e, we can take an institutional *stance*. This is an important engine for social change that enables us to reveal hidden role-based systems that were previously unobservable to us or that were initially conceptualized in other ways. For example, the folk psychological conception of race, gender, and caste-based systems are often essentialized, i.e., thought of as deep quasi-biological kinds. Yet many theorists now analyze them in structural terms, revealing that they are actually hierarchical systems of interlocking social roles in which, for example, *woman* is a structural position defined in subordination to *man* (Haslanger, 2000; Rehbein & Söyler, 2020). Analyzing complex patterns of social behavior in this way can reveal hidden structures, allowing us to then evaluate them and consider how to modify or intentionally subvert them (Hesni, in press). This presents the possibility of recognizing unfair or oppressive systems and considering how to best improve or erase them. In other words, the generative aspect of human institutional reasoning allows us to imagine and pursue genuine institutional alternatives.

The second uniquely-human aspect of social intelligence comes from the ability of institutional structures to create genuinely new domains of behavior, including creating institutional structures where objects can take on roles which then interact with people in powerful new ways. Take for example a *ballot*, the act of *voting*, and a *citizen*. These kinds are not merely *socially constructed* in a narrow sense, such as the sense in which we might say a tool is socially constructed if it required several people to come together to participate in its fabrication. Instead, each of these kinds is *socially constituted* in a stronger sense in which defining them requires invoking social forces, including social roles and the relations between them (what we here refer to as institutions). As discussed extensively by philosophers in the social ontology tradition (Asta, 2018; Searle, 2006; Hindriks & Guala, 2015) and more recently by cognitive scientists (Noyes et al., 2018, 2020a), these entities have a unique ontological status in that their existence is not just brought about by but actively sustained by social forces. An institutional process can render a piece of paper a ballot, a particular act of marking that paper voting, and a particular individual a citizen; a further change in institutional structure can reverse those changes even without affecting any intrinsic or even any physical facts about the object, action, or person. This contrast becomes clear when we consider institutional kinds in comparison to objects like *hammers*, acts like *hammering*, and people who are *carpenters.* The factors which make these things what they are can be defined without directly making reference to social forces (even if, as we alluded to above, social forces may have been involved in initially bringing them about, say by manufacturing them or communally developing a set of practices). Institutional processes might well be involved in regulating the use or manufacture of hammers, the allowable forms of hammering, or the people allowed to perform the role of a carpenter, but however such processes play out we would still consider a certain class of objects designed to deliver force a hammer and the act of driving in a nail hammering. On the other hand, a ballot can really cease to be a ballot, and an action can really cease to be voting, as the direct result of institutional processes.

Foundational work assessing whether these philosophical distinctions map onto everyday psychology suggests that adults and older children assert that changes in institutional practice (such as what most people do and believe) can make an object like a ballot no longer a ballot, or an action like voting no longer voting. By contrast, they assert that this is not the case for objects like hammers, or actions like hammering (Noyes et al. 2018, 2020c, 2023). In other words, for a special class of institutional kinds, people explain their causal powers not via internal features of the objects or agents, but rather because of the way in which those objects or agents relate to patterns of human intention that are themselves positioned within social structures. This work suggests that people come to see certain classes of objects, actions, and human kinds as *institutional* in the relevant sense, raising the possibility that the institutional stance is the cognitive basis of this form of *social reality*, a vital component of the generativity of human social structures.

7. Conclusions and Looking Forward

When comparing human social life to that of other animals, two differences quickly become apparent. First, face-to-face social interactions are qualitatively distinct from those observed in any other animal. This is often thought to reflect people’s ability to build high-resolution models of each other's minds, which then support complex forms of collaboration and communication. Second, human societies have undergone an explosion of social systems that we face in our everyday lives: Coffee shops and restaurants where we interact with strangers, special interest groups and neighborhood associations where we interact with friends and acquaintances, and large-scale formal systems of education and government which structure social life as a whole. If we are correct, these institutions are not a mere byproduct of mentalizing, nor an offshoot of more general non-social forms of intelligence, but rather a direct consequence of a complex framework we have for interpreting—and creating—social behavior through an institutional stance.

While our focus has been on examples for how we skillfully navigate the social world through an institutional stance, there is another powerful source of evidence for these broader phenomena. Consider the phenomenological experience of finding oneself within an institutional stance but realizing that one does not have the knowledge necessarily to fluently navigate it. These experiences range from everyday events like entering an unfamiliar self-serve restaurant to rarer events like traveling to a new country and being ignorant as to how to engage in routine social niceties. In cases like these, feelings of general anxiety about how to behave, or the broader sense of culture shock, plausibly reflect the realization that we are newly amidst a system of rich role-based expectations that we do not fully understand. Indeed, a quite similar analysis has been developed within architectural criticism to explain why spaces like *The Venetian* or *Caesars Palace* in Las Vegas are so successful. They are (admittedly kitschy) physical simulations of other countries, which people can navigate under familiar American consumerist scripts, thus avoiding the alienation and stress of unknown institutional settings (Irazábal, 2007). The anxiety that such experiences produce likely reflects both our ability to detect when an institutional stance is in play (possibly through, e.g., detecting unexpected and not yet fully understood regularities in people's behavior), and our motivation to harness those regularities productively (leading us to watch others more intently to understand what types of roles exist, what characterizes them, and which one we occupy).

Do we need both a mentalistic and an institutional stance? Most domains we evolved to understand are non-reactive: they do not change or react to the cognitive system analyzing it. That is, the laws of physics do not change as a function of how our mind models physics, and faces are not shaped by how we analyze them (at least within shorter evolutionary timescales). By contrast, social behavior is a dynamic reactive system. Whatever system a social group uses to understand each others' behavior also affects social behavior itself (e.g. "looping effects" in which individuals and groups react to how they are conceptualized by others, in turn affecting those conceptualizations; Hacking, 1995). This means that our social cognition is not attempting to model a passive system, but rather an active one that changes as our social cognition evolves, a state of affairs which necessitates interpretive systems of considerable flexibility. We argue that the uniquely dynamic nature of the problem of social cognition produced pressure to co-evolve two complementary systems of analysis: A general mentalistic stance for predicting unconstrained behavior in any context, and an institutional stance for constraining and making behavior predictable when mentalizing fails or when the context is sufficiently familiar to allow leveraging knowledge of recurrent patterns. This latter system emerges as particularly important when reasoning about large-scale cooperative systems and bureaucracies composed primarily of strangers, where role-structure allows one to nonetheless easily navigate in an efficient manner. And while these might have originally emerged as competing solutions to the same problem, the intersection of the two leads us to create complex forms of social intelligence that are unprecedented in any other species.

So far, we have reviewed theoretical positions and empirical literatures that, to varying degrees, support our contention that an institutional stance is a core aspect of human social cognition. In the remainder of this paper, we highlight potential implications of the institutional stance for social cognition more broadly, as well as open questions in which new data would be particularly illuminating.

First, the relative reliance on mentalistic versus institutional understanding is likely shaped by culture. Most directly, certain types of institutional structures might be over- or underrepresented in WEIRD cultures (Henrich et al., 2010), and overreliance on Western social interactions will hinder our understanding of the institutional stance (in perhaps a similar way as plays out in understanding human language; Blasi et al., 2022). It is also possible that the interplay between the two systems varies cross-culturally. For instance, some cultures might have a stronger belief that institutional roles ought to be played out even when they conflict with our own mental states, while other cultures may have an expectation that personal values override institutional roles and failure to do so is morally reprehensible, a phenomenon that we could imagine is linked to aspects of individualism versus collectivism (a dimension we see as analogous to the notion of tight versus loose societies; Gelfand et al., 2017).

Relatedly, the degree to which people rely on the institutional stance might also vary across historical periods. For example, the 18th- and 19th-century regency era of British history—best known in modern times from Jane Austen's novels—was characterized by a heavy focus on social classes, with the upper-classes regulated by a complex set of social roles, each with tight expectations for social behavior across different social events (Erickson, 1986; Le Faye, 2002). Interestingly, Jane Austen's writing itself also reveals how a default institutional interpretation as the backbone of social interaction can give rise to rich and nuanced mentalistic inferences that arise from subtle cues based on how people act, or fail to act, within institutional expectations.

It is also possible that the applicability of an institutional vs mentalistic stance does not depend solely on explanatory power but also on cognitive efficiency. In particular, rich mentalizing about agents is resource intensive and capacity limited. To name just one example, it would be daunting to attempt to track all the minds in a large crowd. Yet, people do readily track large-scale role-based complex behavior. This is particularly evident when watching sports like basketball or football, where what is (to a newcomer) an unintelligible confusion of running bodies becomes (to an expert) an intricate yet intelligible ballet. This happens not via a better leveraging of mental state reasoning but rather via fluency with the internal role structure, which renders each agent's behavior at least in large part predictable and efficiently directed towards higher order goals (a touchdown, say). We suggest that similar dynamics allow us to reason efficiently about other forms of complex multi-party interaction.

If the institutional stance is indeed evolutionary ancient, it is possible that it might enjoy the support of dedicated neural circuitry, similar to the Theory of Mind network (Saxe & Kanwisher, 2003); or it may even be that what we understand to be the ToM network might in fact be the combination of separate mentalizing and institutional frameworks not yet distinguished empirically. It is also an open question whether the institutional stance operates purely within high-level cognition, or whether some aspects could be implemented within perception itself. Recent work has argued that the visual stream has a pathway specialized for social computation, which is both functionally and anatomically different from the ventral and dorsal streams (Pitcher & Ungerleider, 2021; Pitcher, 2021; Grossman, 2021). It is thus possible that visual perception forms a first set of basic representations—agency, goals, and basic social relations (Abassi & Papeo, 2020; Isik et al., 2017; Papeo et al., 2017; Scholl & Tremoulet, 2000)—which can then be interpreted either through mentalizing or through an institutional stance; Fig. 4). But it is also possible that some social computations within vision serve as bottom-up cues for knowing when to deploy an institutional stance. For instance, vision might rapidly extract certain role-like relations from visual input, such as *chaser* and *chased* (Gao et al., 2012; Barret et al., 2005). It is even possible that people exploit principles of how perception works (which did not evolve for the institutional stance) to trigger the institutional stance: For instance, visual synchrony is an important cue within perception for segmenting visual input (Usher & Donnelly, 1998), and groups that should be interpreted through an institutional stance often enact forms of tight synchrony (such as an army marching in unison), which might be a cultural strategy to create a strong visual percept that a set of people ought to be read as a large coordinated institutional group.

One outstanding challenge lies in developing a full-fledged computational theory that can capture the exact nature of the representations and computations that instantiate an institutional stance (akin to frameworks developed for Theory of Mind; Baker et al., 2017; Jara-Ettinger et al., 2020; Lucas et al., 2014; Jern et al., 2021). In our proposal, people have a specialized system for synthesizing and inferring institutional structures—a grammar of institutional structures—but the content embedded in those roles and relations is representationally flexible. As a crude analogy, consider the distinction between abstract linguistic knowledge and the massive space of representations that language manipulates to create complex meaning. One possible candidate representational framework for this problem is probabilistic programming, which enhances symbolic representations with uncertainty, and has been found to be a powerful framework to explain high-level cognition (Lake et al., 2017; Tenenbaum et al., 2011). Computational work will be necessary to shed light on these possibilities.

We have focused primarily on the power and utility of an institutional stance as a lens through which to do social cognition, but applying it to others likely comes with a dark side. Consider the frequency of attempting to justify morally unjustifiable acts via appeal to the mere enactment of institutional roles, as perhaps most famously in the trial of Nazi officer Adolf Eichmann, who defended himself via the claim that he was “just” executing a role (Arendt, 2006). More generally, if institutional and mentalistic stances sometimes trade off against one another, then applying an institutional stance might reduce the extent to which we represent the mental and emotional life of others, thereby reducing empathy and stripping others of their humanity (Haslam & Loughnan, 2014; Haslam, 2006; Harris & Fiske, 2006). If so, the links between the institutional stance and dehumanization might run deep, from rather mundane cases of “everyday” dehumanization such as engaging with your barista instrumentally to more severe and consequential forms of dehumanization that spur serious conflict. If so, evolution’s attempt to help us understand each other might, in some cases, prevent us from seeing who we really are.

But the institutional stance is also a source of great human potential. Questioning existing categories and subverting long-standing exploitative patterns seems to require the cognitive tools we have outlined here. Thus, the same psychological forces which can reduce people to cogs in institutional structures also offer the possibility of envisioning and enacting new and more just futures. The institutional stance gives us the tools to analyze rather than blindly accept institutional structures, to recognize unjust and exploitative systems, and to imagine possibilities for positive change. In these cases our humanity most shines when we consider the roles we play.

Acknowledgments

We thank Paul Bloom, Eric Mandelbaum, Alexander Noyes, Alexander Paunov, Katie McAuliffe, Kate Richie, and Denis Tatone for helpful conversations. This work was supported by NSF awards BCS-2045778 and IIS-2106690. The authors declare no conflicts of interest.

Funding Statement

This work was supported by NSF award BCS-2045778.

Conflict of Interest Statement

The authors declare no conflicts of interest.

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